Power factor correction — MDB-1

Villa Complex 415V LV Network · REV — · 15 October 2026

Capacitor bank 175 kvar installed = 7 × 25 kvar, 7 % detuned — at this load 7 steps (175 kvar) switched in: PF 0.80 lagging → 0.954 lagging

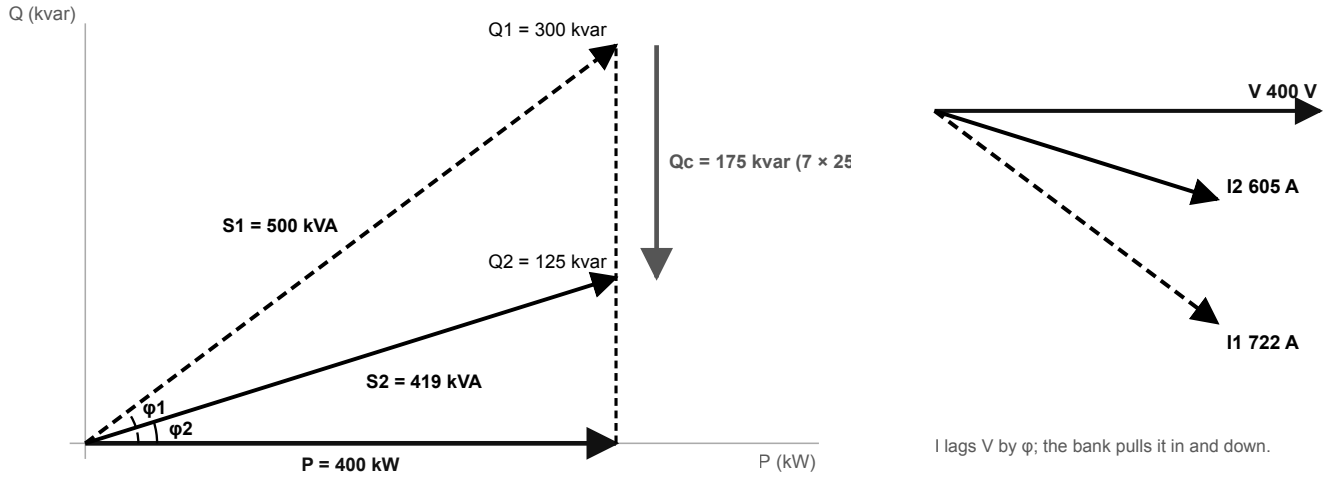
Input data

Input	Measured kW and power factor
System	400 V, 3-phase, 50 Hz
Target PF	0.95
Step size	25 kvar
Harmonics	Some non-linear load (VFDs, UPS, LED)

Method

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Q = P × tan φ1 = 400 × tan(acos 0.8) = 300.0 kvar
Qc = P × (tan φ1 − tan φ2) = 400.0 × (0.750 − 0.329) = 168.5 kvar
Installed 175 kvar = 7 × 25 kvar; switched in at this load: 7 steps = 175 kvar (fewest steps reaching the target without going leading)
Q2 = Q1 − switched kvar = 300.0 − 175 = 125.0 kvar → PF2 = P ÷ √(P² + Q2²)
Bank: next standard size in whole 25 kvar steps = 175 kvar installed; the relay switches whole steps — the fewest that reach the target without going leading (else the most that stay lagging): 7 × 25 = 175 kvar → Q2 = Q1 − 175 = 125.0 kvar, PF2 = P ÷ √(P² + Q2²) = 0.954 lagging. All after-values below are for this switching state.
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Power triangle — before (dashed) and after (solid), to scale



φ1 = 36.9° · PF 0.800 lagging (before) **φ2 = 17.4°** · PF 0.954 lagging (after)

Before and after

	Before	After
Active power P	400.0 kW	400.0 kW
Reactive power Q	300.0 kvar	125.0 kvar
Apparent power S	500.0 kVA	419.1 kVA
Power factor	0.800 lagging	0.954 lagging
Phase angle φ = atan2(Q, P)	36.9°	17.4°
Current at 400 V	722 A	605 A

kVA released **80.9 kVA** · current down **16 %** · I²R losses down **30 %**

Bank selection

Required	168.5 kvar
Selected (installed)	175 kvar, automatic, 7 steps × 25 kvar, with power factor relay (CT on the incomer)
Switched in at this load	7 steps = 175 kvar → PF 0.954
Detuning	Some non-linear load (VFDs, UPS, LED): 7 % detuned reactors (189 Hz) to avoid resonance
Capacitor rated voltage	460 V
Bank current	253 A
Breaker	400 A MCCB (≥ 1.43 × bank current, IEC 60831)

Other targets

Target PF	Required kvar	Standard bank	Switched in · PF achieved with 25 kvar steps
0.90	106.3	125	125 kvar · 0.916
0.95	168.5	175	175 kvar · 0.954
0.98	218.8	250	225 kvar · 0.983
1.00	300.0	300	300 kvar · 1.000

Notes

- **PF 0.80 is below the authority's 0.90 minimum (Rules.xlsx)**

Minimum power factor required: 0.90. Detuning recommendations are indicative — confirm with a harmonic survey where VFDs, UPS or large LED loads are present.